



PREDICTIVE WNBA VIABILITY

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INTRODUCTION

The goal of this project is to build a predictive model that estimates professional WNBA viability of current NCAA players using decision tree modeling and ensemble learning.

The professional and collegiate game stats of WNBA players, both current and historic, were compiled and used to train the model. The model was segmented by three positions: guard, forward, and center.

Player viability was defined using five tiers, which were determined by a composite score derived from a weighted combination of professional stats performance metrics, longevity in the league, and other key success indicators.

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NCAA®



WNBA

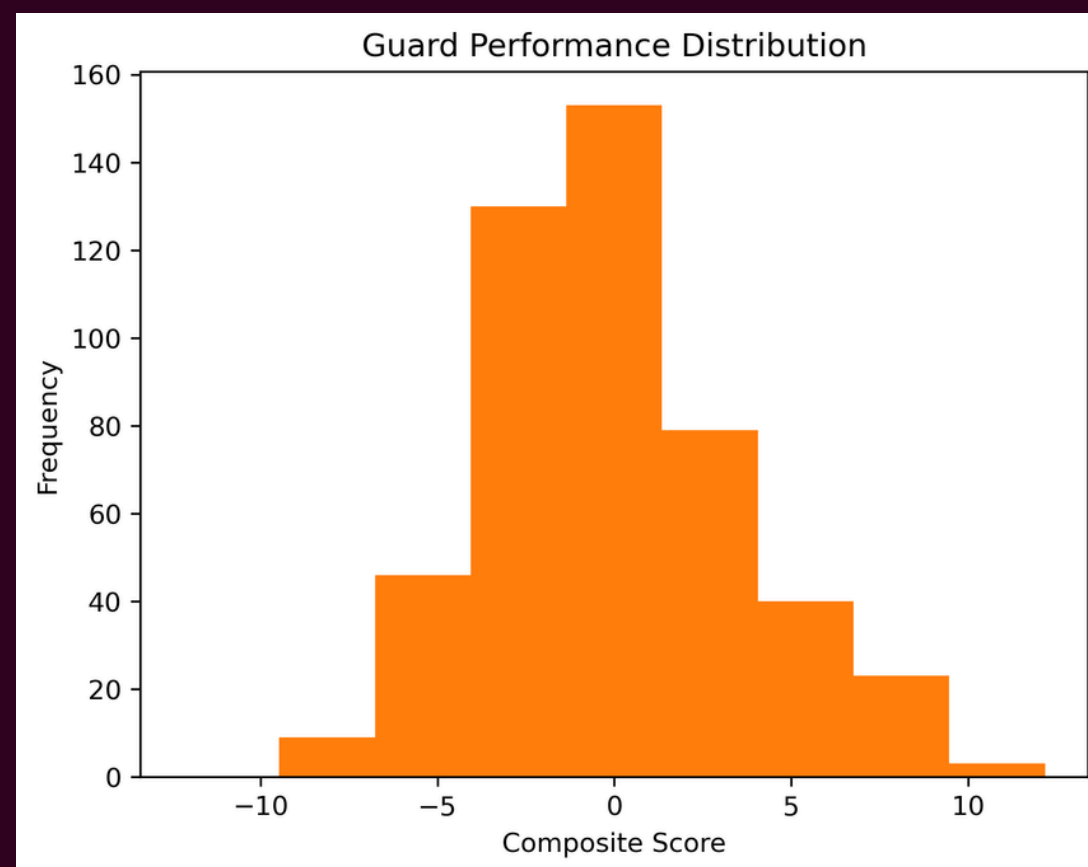


DATA SOURCES

Type	Source	Method	Records	Date Range	Notes
WNBA player box statistics	sportsdataverse wehoop-wnba-stats-data	Download (Parquet)	116,701	2002-2025	Full game stats per player. Repo is aggregations of espn.com box score game data.
WNBA player box statistics	basketball-reference.com	Web Scrape	29,983	1997-2003	Full game stats per player.
NCAA player box statistics	espn.com	API	1,500	2009-Present	Season averages and totals.
NCAA player box statistics	sports-reference.com	Web Scrape	2,037	1981-Present	Season averages and totals. Source was used to supplement missing statistics from espn.com
Draft Order	wnba.com	Web Scrape	1,258	1997-Present	Draft order for players. Includes allocation draftees assigned as draft position 1.



COMPOSITE SCORE



Features

39 features were used



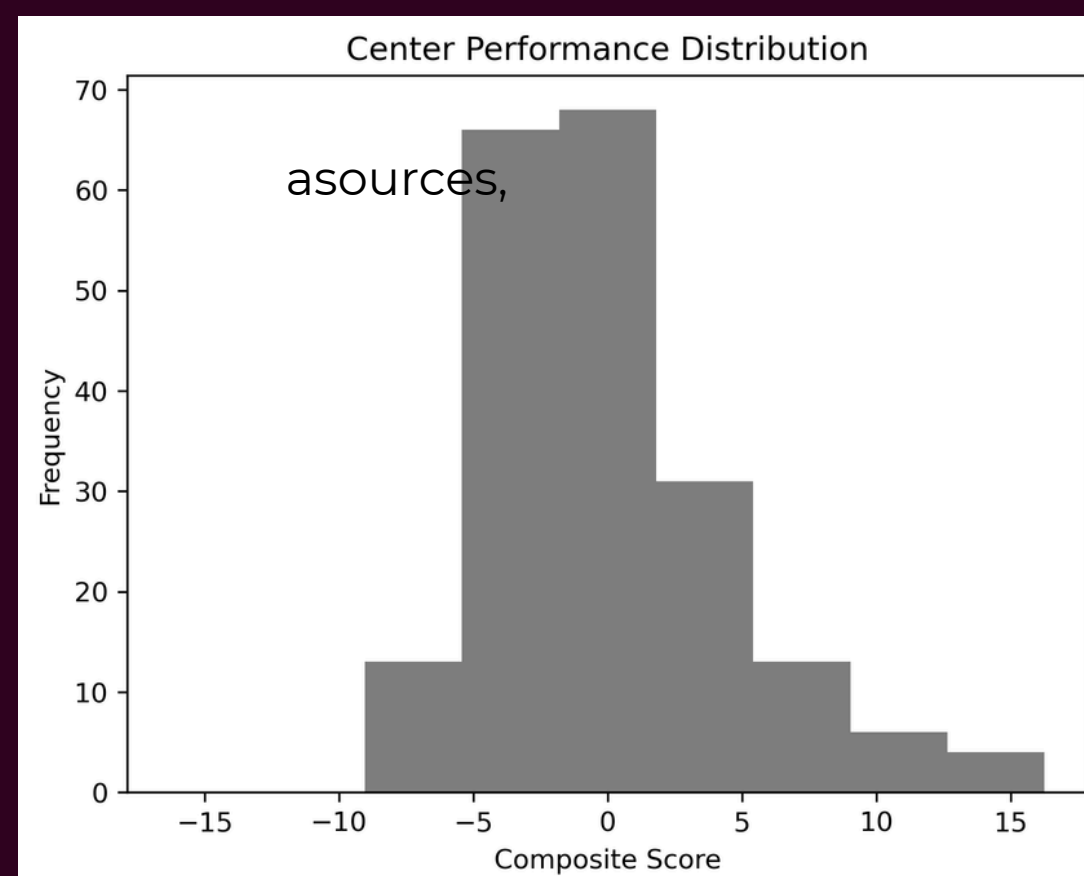
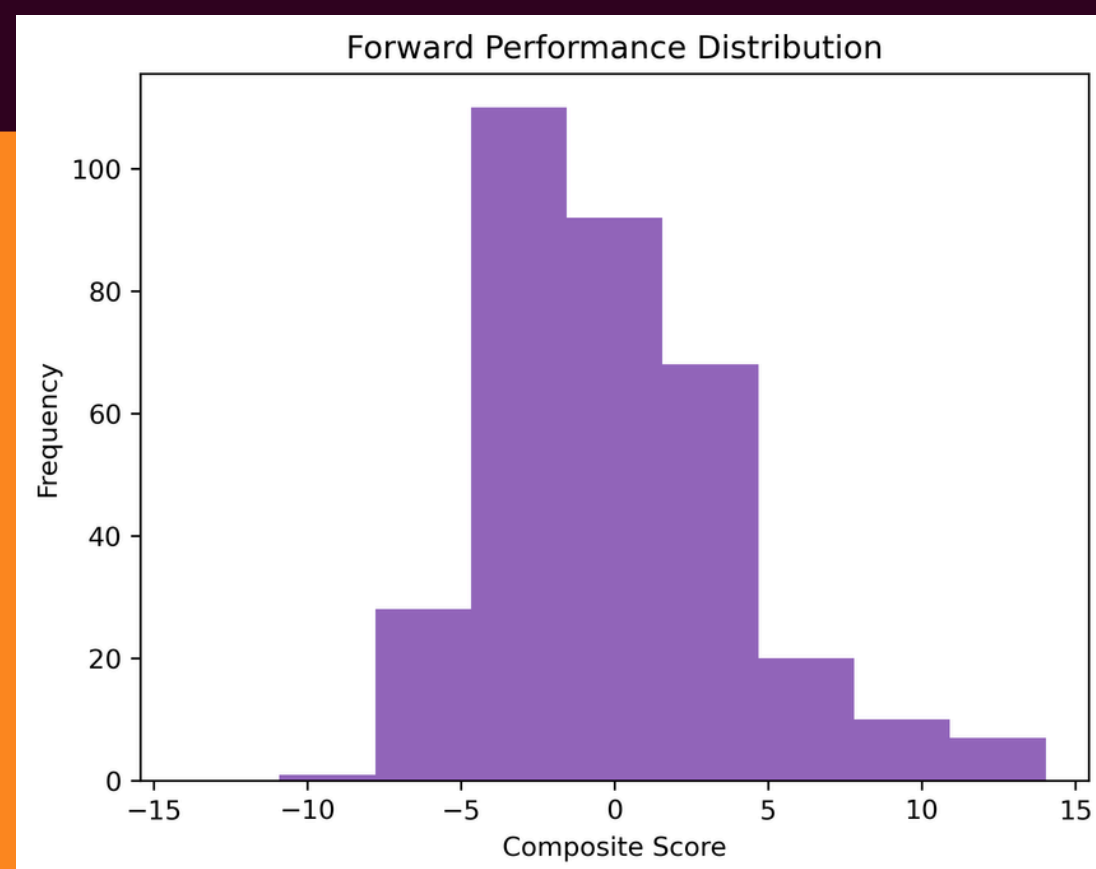
Weighted Score

Scoring + Efficiency + Playmaking + Defense -
Negative Performance + Position(x) + Longevity(x)



Tier Breakdown

Elite - top 10%, Core Performers - 25%, Bench
Backups - 25%, Roster Fillers - 25%, Not-Viable/Wash -
bottom 15%



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FINAL SAMPLE

Type	Sample	Notes
Total Players	1,020	Total number of players that were present in both the WNBA and NCAA datasets after processing. Guards = 483, Forwards = 336, Centers = 201
Composite Score Features	39	assist_turnover_ratio, assists, blocks, defensive_rebounds, field_goal_percentage, field_goals_attempted, field_goals_made, fouls, free_throw_percentage, free_throws_attempted, free_throws_made, minutes, offensive_rebounds, points, rebounds, steal_turnover_ratio, steals, three_point_field_goal_percentage, three_point_field_goals_attempted, three_point_field_goals_made, turnovers, true_shooting_percentage, three_point_attempt_rate, free_throw_attempt_rate, usage_percentage, points_per_minute, rebounds_per_minute, blocks_per_minute, steals_per_minute, turnovers_per_minute, drive_aggression, perimeter_shooting, rebound_share, total_minutes, games_started, total_games, game_availability_percentage, total_seasons, draft_pick Features were grouped according to 7 categories listed in composite formula used to derive performance tiers. Only WNBA data was used in this calculation; NCAA data was used exclusively to train and test models.



DECISION TREE

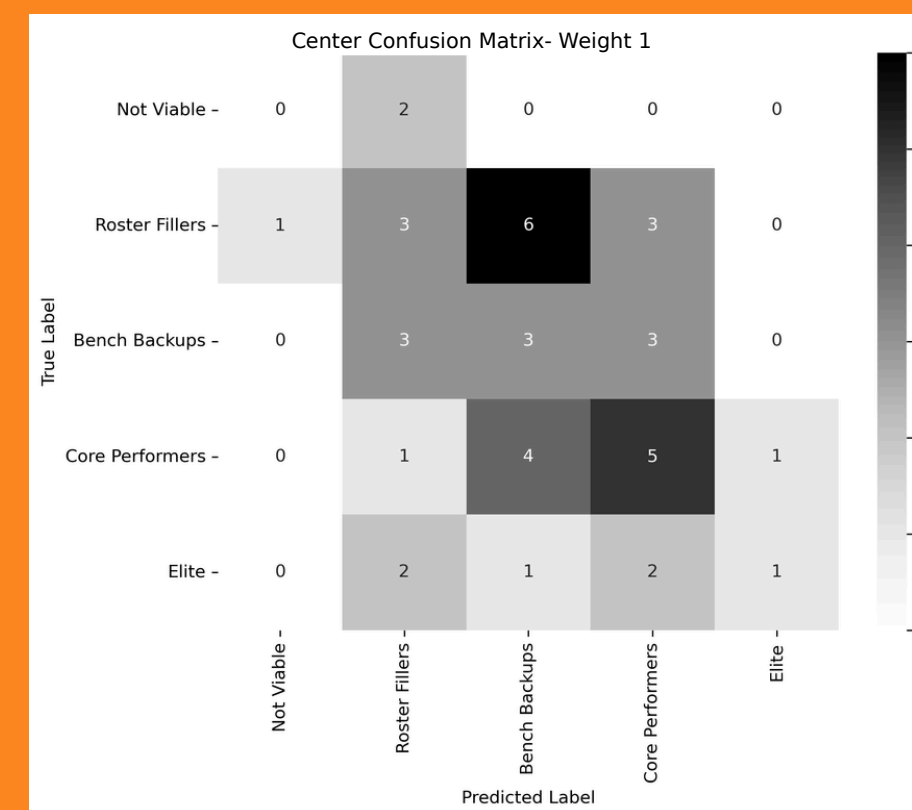
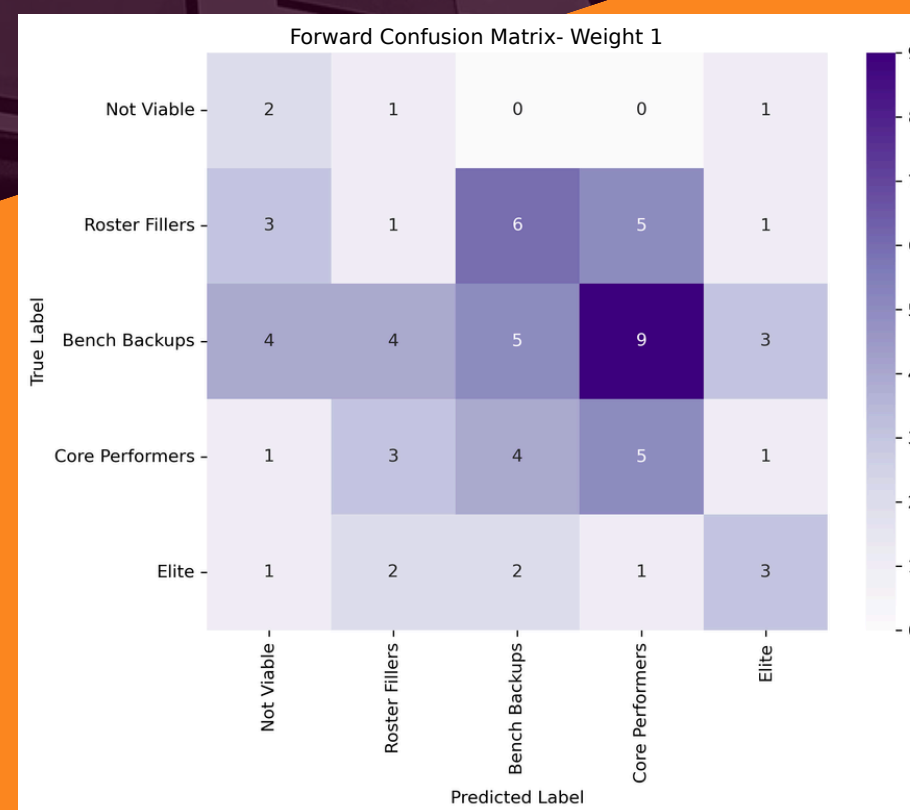
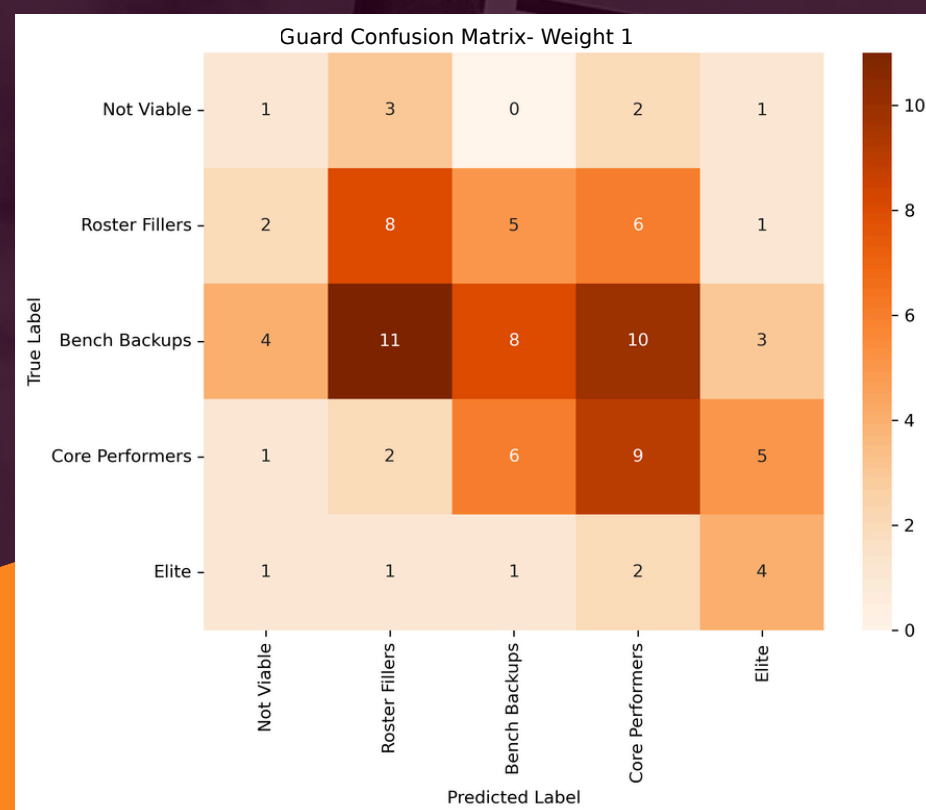
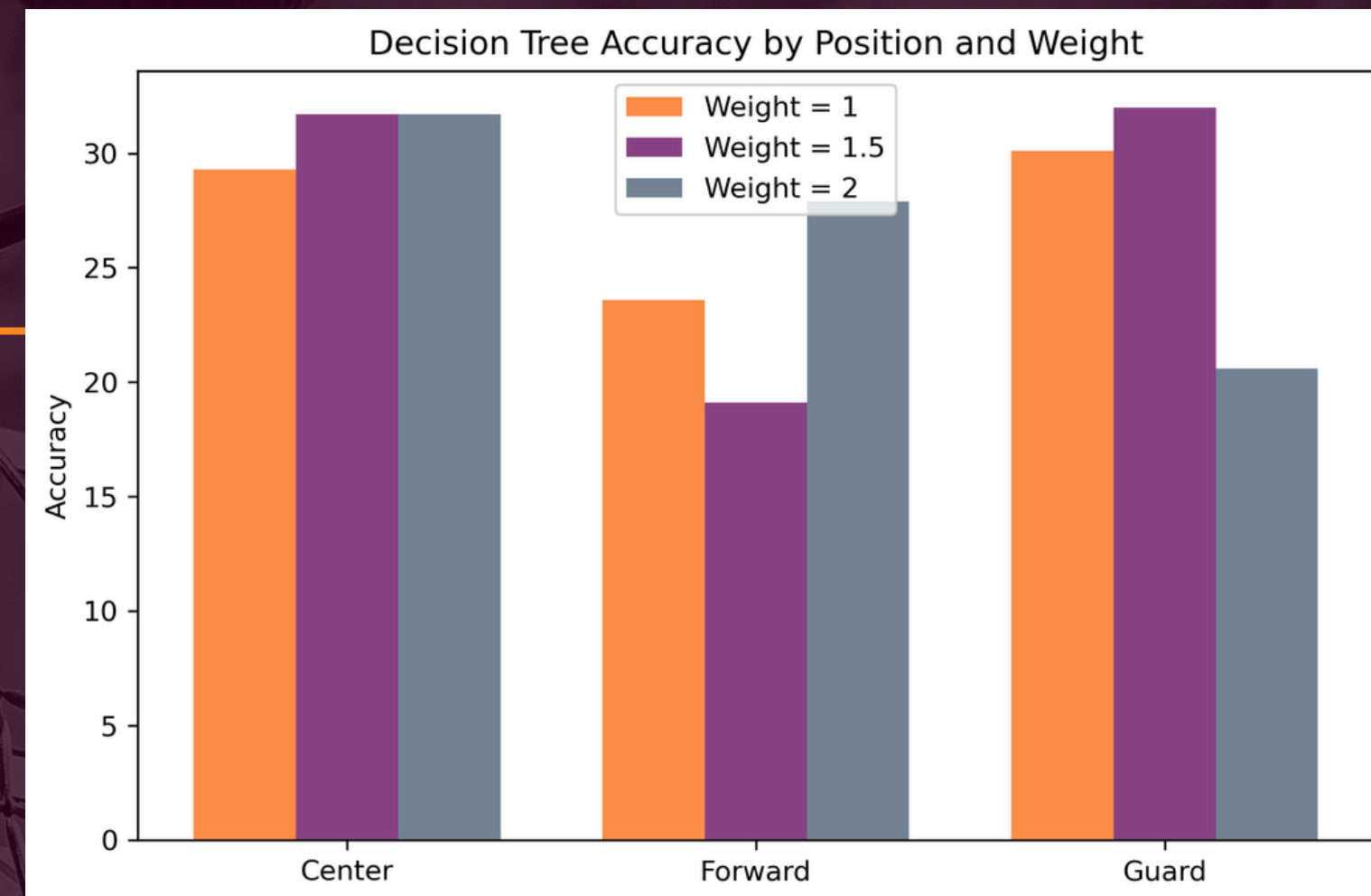
Performance varied across feature weightings but there was no consistent improvement pattern.

Accuracy

Weight 1: 29% Center, 24% Forward, 30% Guard

Weight 1.5: 32% Center, 19% Forward, 32% Guard

Weight 2: 32% Center, 28% Forward, 21% Guard





ENSEMBLE

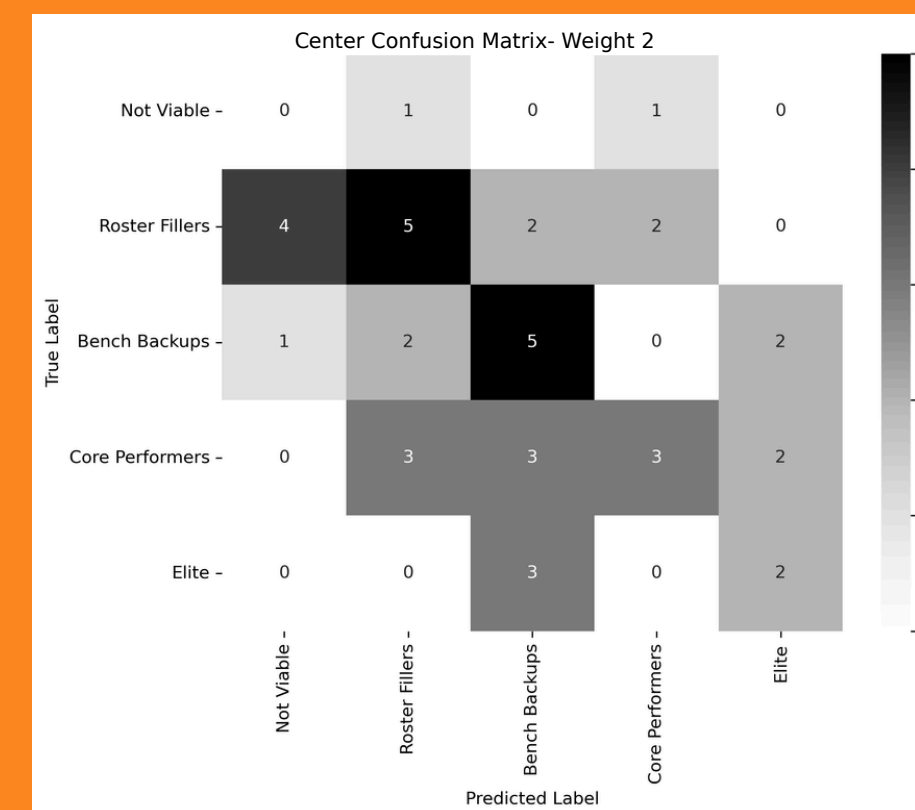
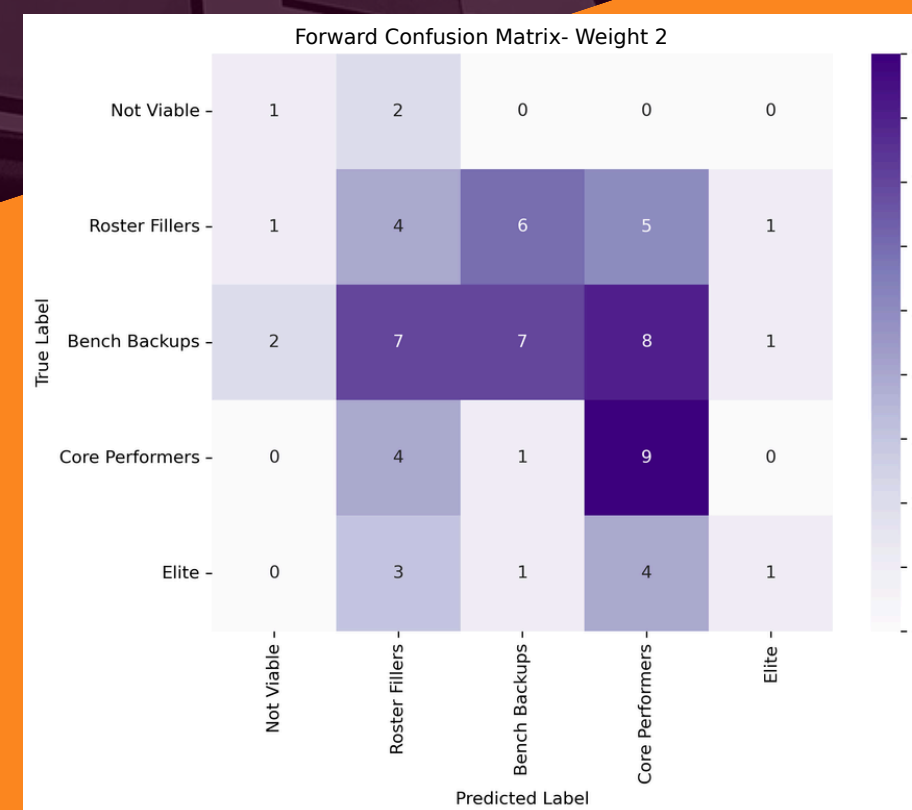
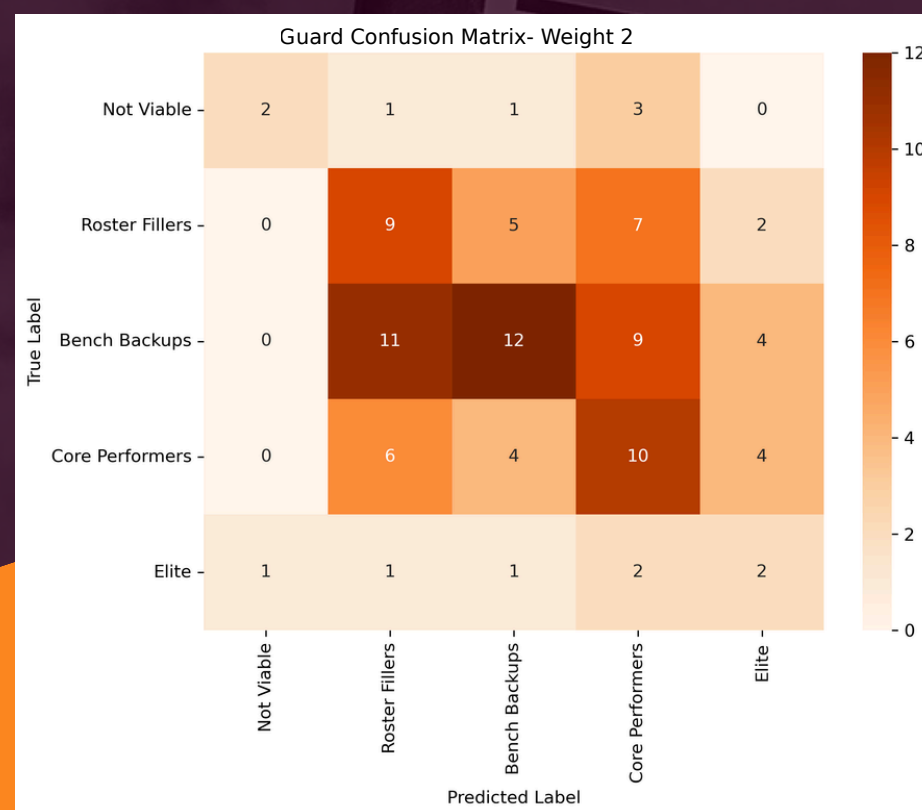
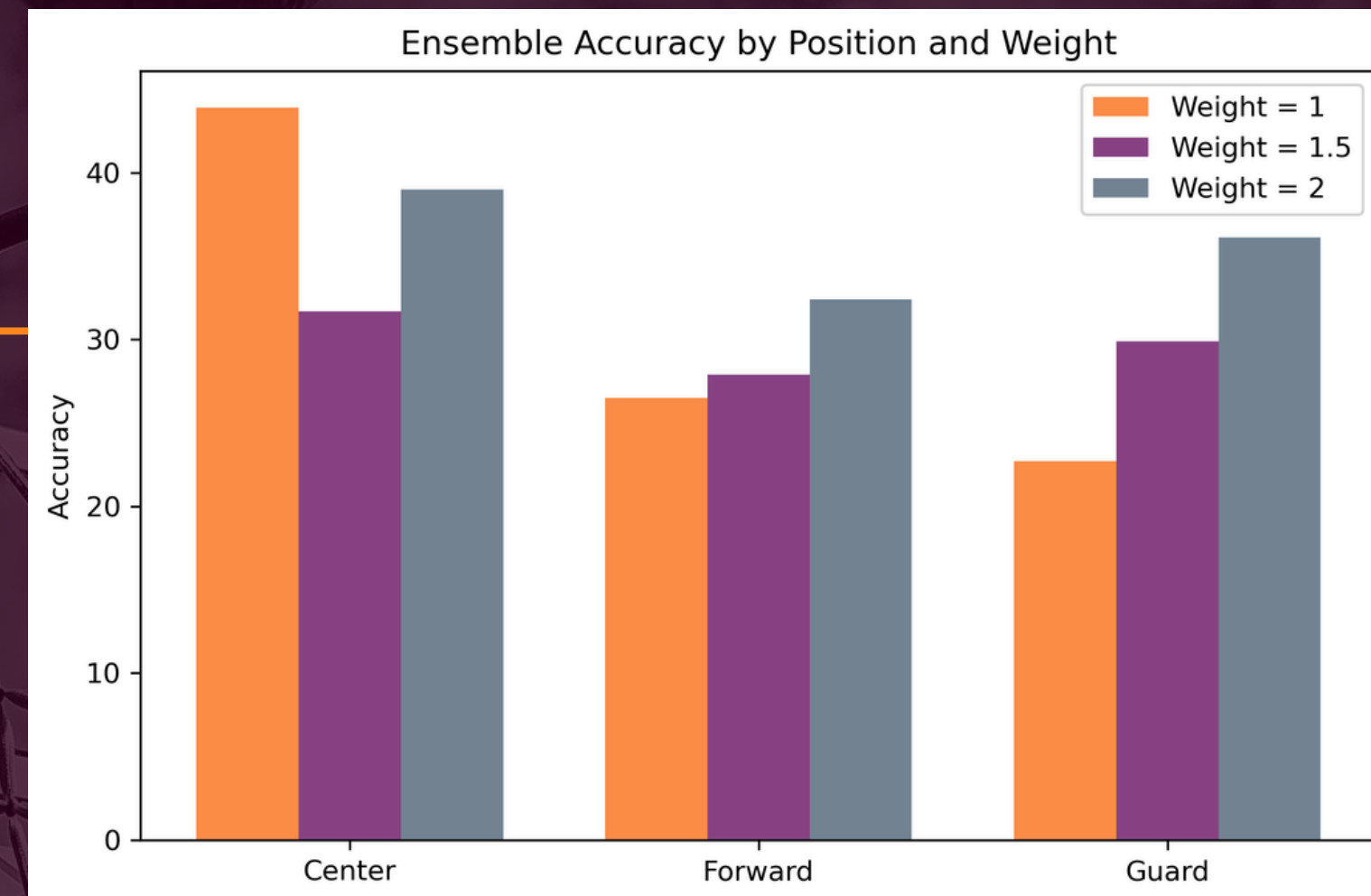
Performance depended on feature weighting with different positions achieving higher accuracy based on selected weight.

Accuracy

Weight 1: 44% Center, 27% Forward, 23% Guard

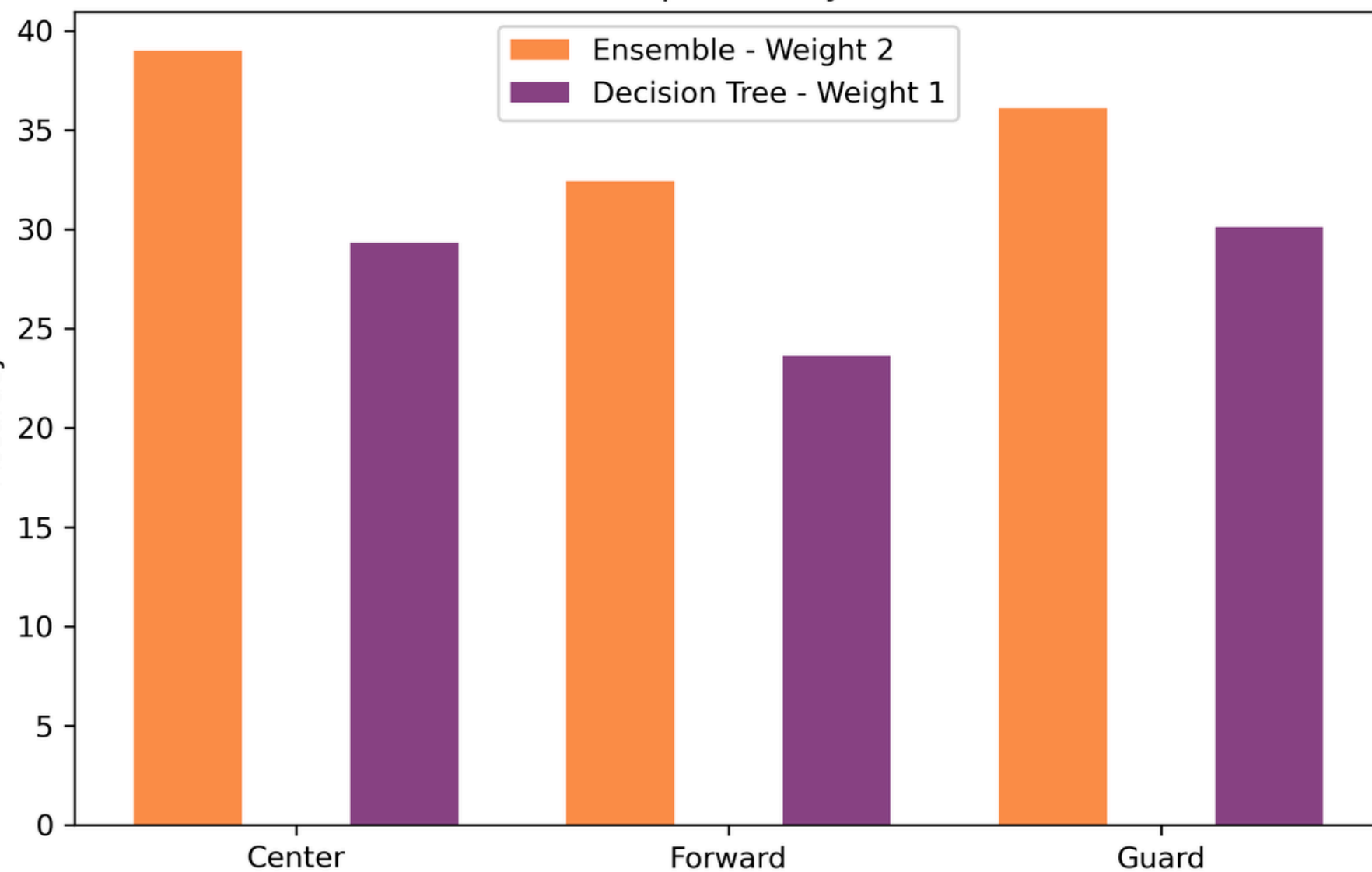
Weight 1.5: 32% Center, 28% Forward, 30% Guard

Weight 2: 39% Center, 32% Forward, 36% Guard





Model Comparison by Position



Ensemble models overall had a higher accuracy ceiling - 43.9% depending on weighting

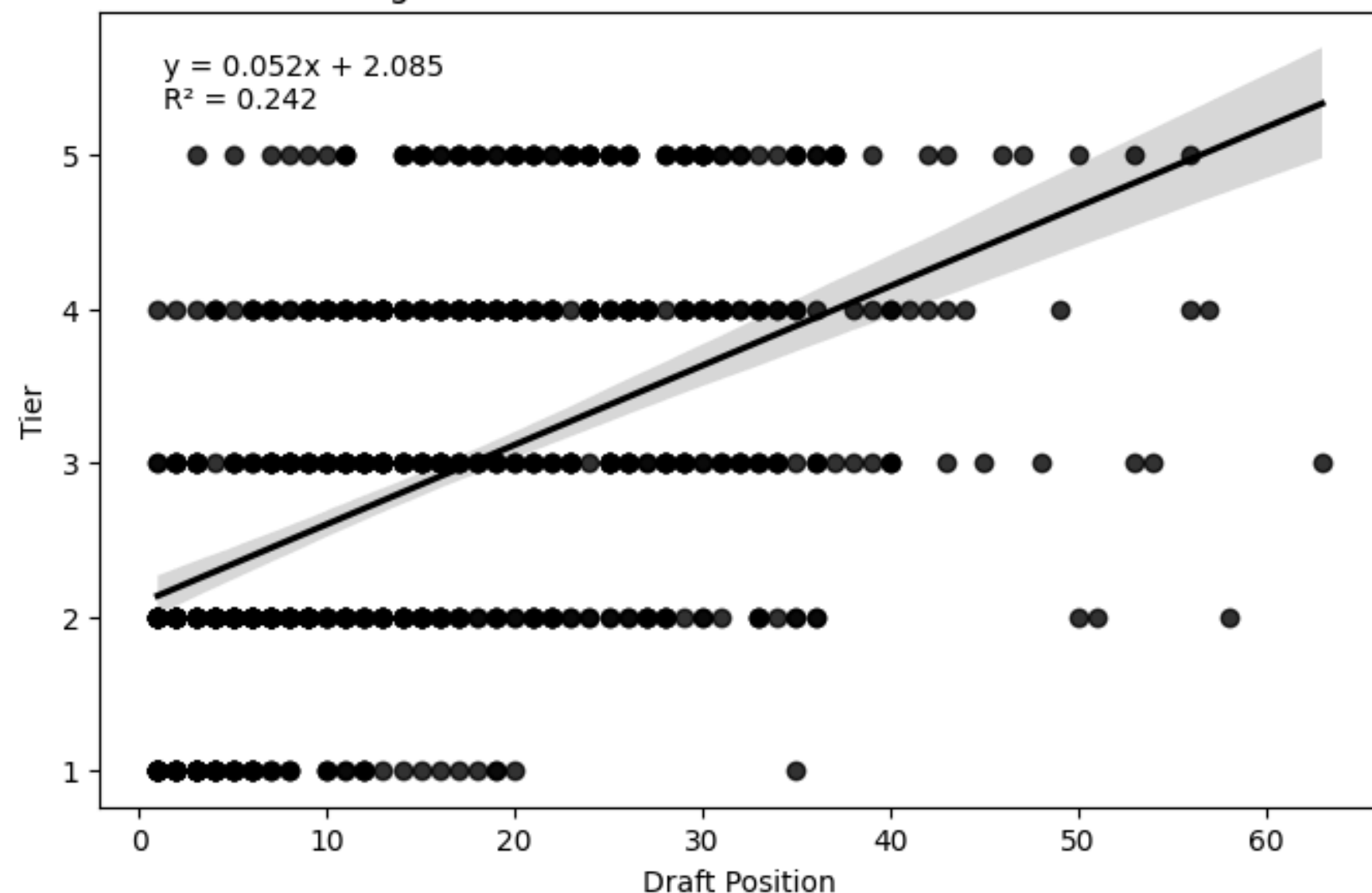
SUCCESS RATE

The ensemble models outperformed decision trees and were sensitive to feature weighting. Centers were the most accurately predicted position overall while forwards were the most variable. Both models were better at identifying the middle 3 tiers of players than the elite and not viable categories.

- ✓ **Center - Highest Accuracy Ceiling****44%**
- ✓ **Center - Most Predictable****35%**
- ✓ **Forward - Least Predictable****26%**



Regression - Draft Position x Performance Tier



Mean Square Error = 1.146

DRAFT BASELINE

Regression examining the predictive power of draft position on performance tier explained 24.2% of the variance in the data with no weighting applied, not adjusted for specific positions. The predictions have an average error of one performance tier. Compared to this baseline, the ensemble model achieves up to 44% accuracy, indicating the feature set included in the model have predictive power that exceeds draft position.



The ensemble model has potential to support teams with draft selection by identifying performance patterns within the NCAA statistics that are indicative of WNBA success.



CHALLENGES

- Cleaning the data
 - Inconsistent data availability between the WNBA and NCAA limited the features that could be used.
- Disappearing data
 - Some data sources became unavailable during scraping, requiring adjustments.
- Scope Creep
 - A large number of variables were explored but not all contributed meaningfully to the models.





THANK YOU



Q&A

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